YAKEEN

NEET 2.0 2027

Physics | Lecture - 2

Basic Maths and Calculus

By- Manish Raj (MR Sir)



Topics to be covered

1

feel of Trigonometry



Telegram chu

Question

Two towers are separated by a horizontal distance D. From the top of the first tower (height h_1), the angle of depression to the top of the second tower (height h_2) is θ.

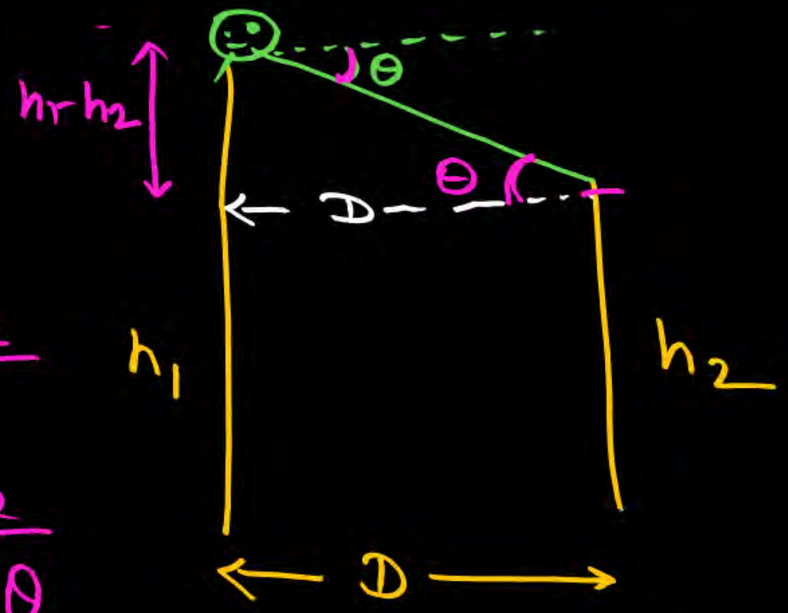
Which relation correctly gives the horizontal separation D in terms of h_1 . h_2, θ?

1. $D = \dfrac{h_1 + h_2}{\tan\theta}$

2. $D = \dfrac{h_1 - h_2}{\tan\theta}$

3. $D = \dfrac{h_2 - h_1}{\tan\theta}$

4. $D = \tan\theta \,.(h_1 + h_2)$

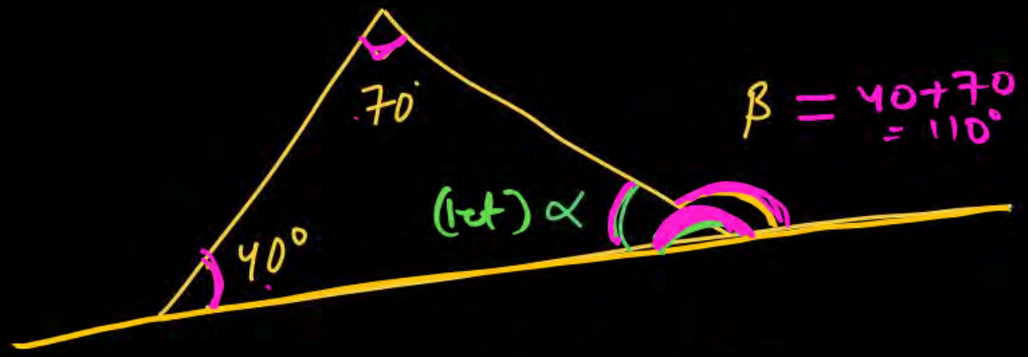


$\tan\theta = \dfrac{h_1 - h_2}{D}$

$D = \dfrac{h_1 - h_2}{\tan\theta}$

H/W Copy

① find β



$\beta = 40+70 = 110°$

$70° + 40° + \alpha = 180$

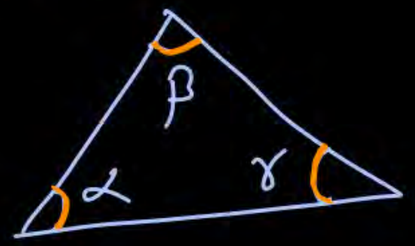
$\Rightarrow \alpha = 180 - 110 = \underline{70°}$

$\beta = 180 - \alpha = 180 - (70)$

$= \underline{110°}$

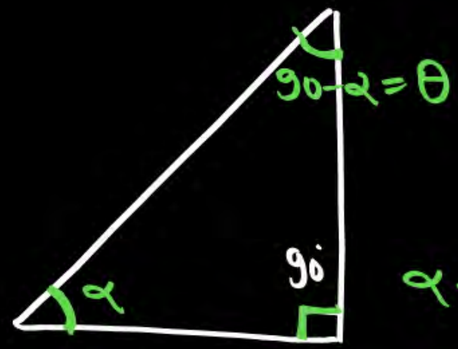
Concept:-

#



$\alpha + \beta + \gamma = 180°$ ✓

#



$\alpha + 90 + \theta = 180$

$\theta = 180 - 90 - \alpha$

$= \underline{90 - \alpha}$

Box

rad into degree.

π rad = 180°

Question

some example
Copy in note

H/W

Convert following radian into degree?

(i) $\frac{\pi}{2}$ rad $= \frac{180^\circ}{2} = 90^\circ$

(ii) $\frac{\pi}{4}$ rad $= \frac{180}{4} = 45^\circ$

(iii) $\frac{\pi}{3}$ rad $= \frac{180}{3} = 60^\circ$

(iv) $\frac{5\pi}{6}$ rad $= \frac{5}{6} \times 180^{30} = 150^\circ$

(v) $\frac{2\pi}{3}$ rad $= \frac{2}{3} \times 180^{60} = 120^\circ$

(vi) $\frac{4\pi}{3}$ rad $= \frac{4}{3} \times 180^{60} = 240$

(vii) $\frac{3\pi}{5}$ rad $= \frac{3}{5} \times 180^{\times 36} = 108^\circ$ ✓

(viii) 5π rad $= 5 \times 180 = 900^\circ$

(ix) $\frac{2\pi}{5}$ rad $= \frac{2}{5}(180)$ ✓

(x) $\frac{7\pi}{12}$ rad $= \frac{7}{12} \times 180^{15} = 105^\circ$

(xi) $\frac{5}{2}\pi$ rad $= \frac{5}{2} \times 180 \cdot 90 = 450^\circ$

(xii) $3\ (\pi$ rad$) = 3 \times 180 = 540^\circ$

(xiii) $\frac{\pi}{6}$ rad $= \frac{180}{6} = 30^\circ$

Question

copy some example (HW)

Convert following into radian

(i) $45^\circ = 45\left(\frac{\pi \text{ rad}}{180}\right) = \frac{\pi}{4}$ rad

(ii) $60^\circ = \frac{60 \times \pi \text{ rad}}{180} = \frac{\pi}{3}$ rad

(iii) $120^\circ = \frac{120 \times \pi \text{ rad}}{180} = \frac{2\pi}{3}$ rad

(iv) $30^\circ = \frac{\pi}{6}$ rad

(v) $150^\circ = 150 \times \frac{\pi}{180} = \frac{5\pi}{6}$

(vi) $90^\circ = \frac{\pi}{2}$ rad

(vii) $\pi^\circ = \pi \times 1^\circ = \pi \times \frac{\pi \text{ rad}}{180} = \frac{\pi^2}{180}$ rad $= \frac{10}{180} = \frac{1}{18}$ rad.

(viii) $270^\circ = \frac{3}{2}\pi$ rad

(ix) $135^\circ = \frac{3\pi}{4}$ rad.

$180^\circ = \pi$ rad

$1^\circ = \frac{\pi \text{ rad}}{180^\circ}$

$\pi = 3.14$

4° $(3.14)^\circ$ $(8.8)^\circ$ $\pi^2 = (3.14)^2 \cong 10$

$\pi = 180^\circ$ ✗

$\pi \times$ rad $= 180^\circ$ ✓

Question HW No copy

Find value of sin (53°) × cosec (53°) = 1 ✓

$\sin(\theta) \times \operatorname{cosec}(\theta) = 1$

Question

M/w No Cop

Find value of sin (180°) cos (135°) tan (120°)

$\tan(120^\circ) = -\sqrt{3}$

$\cos(135^\circ) = -\frac{1}{\sqrt{2}}$

$\sin(180^\circ) = 0$

Table
Revision 0

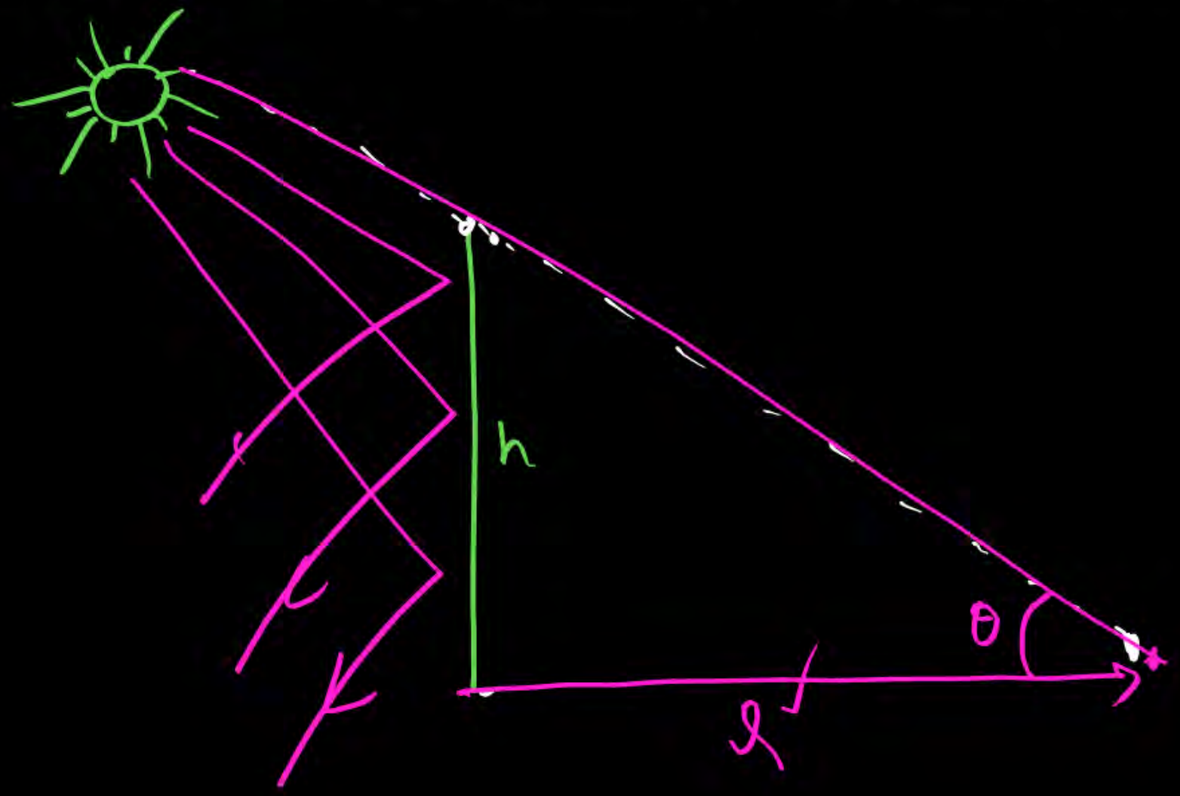
Question

H/w No copy

A vertical pole of height h casts a shadow of length l when the Sun is at an angle θ above the horizon.

Which of the following expresses the shadow length l in terms of h and θ?

1. $l = h \,.\, \tan\theta$
2. $l = \dfrac{h}{\tan\theta}$
3. $l = h \,.\, \sin\theta$
4. $l = \dfrac{h}{\sin\theta}$



$\tan\theta = \frac{h}{l}$

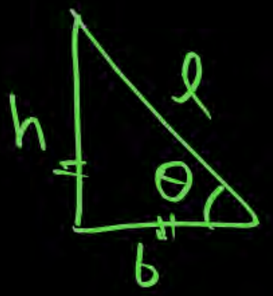
$l = \frac{h}{\tan\theta}$

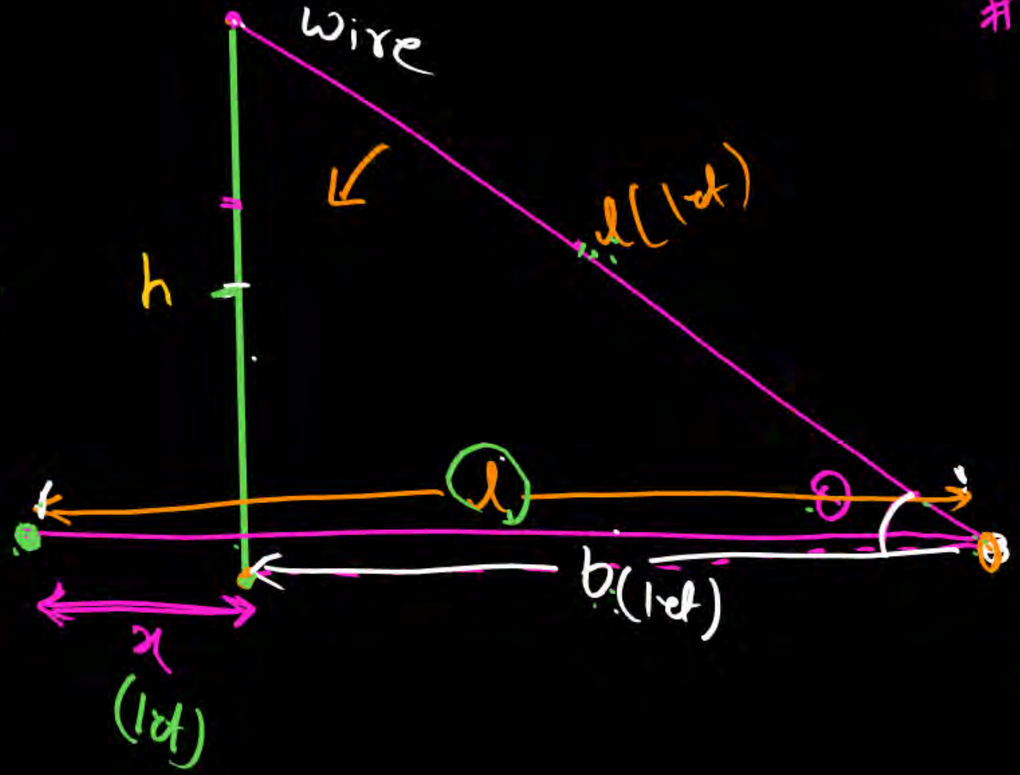
Question

H/w Copy ghar

A pole of height h has a wire attached from the top to the ground at an angle θ. If the wire breaks and falls to the ground without changing length, find the horizontal distance x from the base of the pole where the wire touches the ground.

1. $x = h \cdot \cot\theta$
2. $x = h \cdot \tan\theta$
3. $x = h \cdot \sin\theta$
4. $x = h \cdot \sec\theta$





$\sin\theta = \frac{h}{l}$

$l = \frac{h}{\sin\theta} = h\operatorname{cosec}\theta$

$\tan\theta = \frac{h}{b}$

$b = \frac{h}{\tan\theta} = h\cot\theta$

$x = l - b = h\operatorname{cosec}\theta - h\cot\theta$

$= h[\operatorname{cosec}\theta - \cot\theta]$

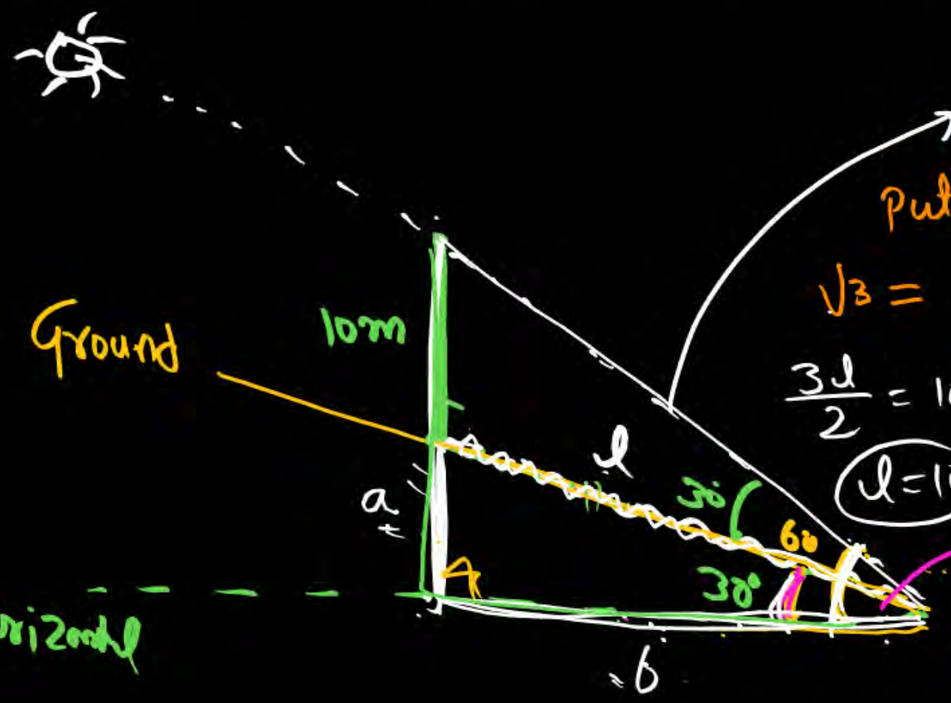
Question

$\sqrt{3} \times \sqrt{3} = 3$ H/W copy

exiP

A vertical pole of height $h = 10$ m stands on ground that slopes upwards at a constant angle $\alpha = 30°$ with the horizontal. If the sun's angle of elevation above the horizontal is $\theta = 60°$, what is the length of the shadow cast by the pole on the sloping ground?

1. 5 m
2. 10 m
3. $10\sqrt{3}$ m
4. $\frac{10}{\sqrt{3}-1}$ m



large triang. $\tan 60° = \frac{10+a}{b}$

$\sqrt{3} = \frac{10+a}{b}$ - ①

Put a & b ← $\sqrt{3} = \frac{10 + l/2}{\frac{\sqrt{3}l}{2}}$

$\frac{3l}{2} = 10 + \frac{l}{2}$

$l = 10$

$\sin 30° = \frac{a}{l} \Rightarrow a = l \sin 30°$

$a = \frac{l}{2}$

$\cos 30° = \frac{b}{l} \Rightarrow b = l \cos 30°$

$b = \frac{l\sqrt{3}}{2}$

COPY

$1\,m = 100\,cm$

$1\,fm = 10^{-15}\,m$

$1\,pm = 10^{-12}\,m$

$1\,nm = 10^{-9}\,m$

$1\,\mu m = 10^{-6}\,m$

$1\,mm = 10^{-3}\,m$

$1\,cm = 10^{-2}\,m$

$1\,km = 10^{3}\,m$

$1\,Mm = 10^{6}\,m$

$1\,A^{\circ} = 10^{-10}\,m$

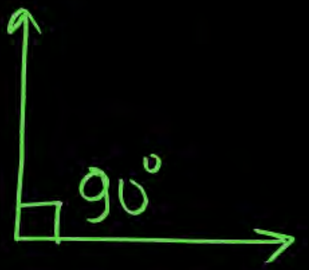
COPY

- Parallel line



$\theta = 0°$

- orthogonal line (Perpendicular)



- Antiparallel line



- Complementary Angle

$\alpha + \beta = 90°$

"$\alpha = 90 - \beta$"

α & β are complementary angle.

- Acute Angle



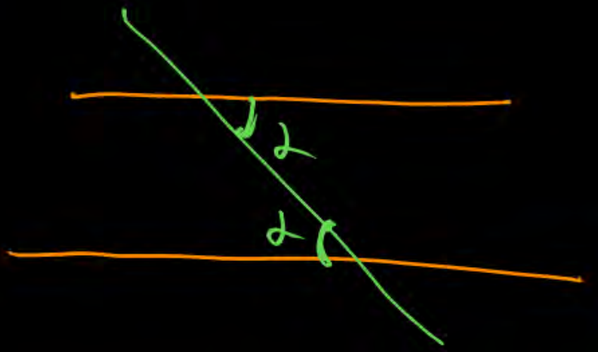
$0° \leq \theta < 90°$

- obtuse Angle



$180° \geq \alpha > 90°$

- Alternate Angle



Copy

Question

If $y = \frac{\sin\theta}{\theta}$ then find value of y if $\theta = 30°$.

Copy

Soln

$$y = \frac{\sin(\theta)}{(\theta)}$$

if $\theta = 30° = \frac{\pi}{6}$ rad

$$y = \frac{\sin(30)}{30°}$$

$$y = \frac{\sin(30)}{\frac{\pi}{6}} = \frac{\frac{1}{2}}{\frac{\pi}{6}} = \frac{1}{2 \times \frac{\pi}{6}_3} = \frac{3}{\pi}$$

$$\frac{a/b}{c/d} = \frac{ad}{bc}$$

copy

Important Trigonometric Relation :**

A & B are Angle

$$\sin(A+B) = \sin A \cdot \cos B + \sin B \cdot \cos(A)$$

$$\sin(A-B) = \sin A \cdot \cos B - \sin B \cos A$$

$$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \cdot \tan B}$$

$$\tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A \cdot \tan B}$$

$\cos(A+B) = \cos A \cdot \cos B - \sin A \cdot \sin B$

$\cos(A-B) = \cos A \cdot \cos B + \sin(A)\sin(B)$

10-times likh ke
Yaad Paye me Practic Karna hai.

@p`

$\sin(A+B) = \sin A \cdot \cos B + \sin B \cdot \cos A$

If $A = B = \theta$ [Both are equal angle]

$$\sin(2\theta) = \sin\theta \cdot \cos\theta + \sin\theta \cdot \cos\theta$$

*** $\boxed{\sin(2\theta) = 2\sin\theta \cdot \cos\theta}$

→ Half angle formula

$2\theta = \alpha$ (let)

$$\sin(\alpha) = 2\sin(\alpha/2)\cos(\alpha/2)$$

COPY

$\cos(A+B) = \cos A \cdot \cos B - \sin A \cdot \sin B$

$A = B = \theta$ (माना)

Soln

$\cos(2\theta) = \cos\theta \times \cos\theta - \sin\theta \cdot \sin\theta$

$\cos(2\theta) = \cos^2\theta - \sin^2\theta$

$\cos(2\theta) = 1 - \sin^2\theta - \sin^2\theta$

$\cos(2\theta) = 1 - 2\sin^2\theta$

$2\sin^2\theta = 1 - \cos(2\theta)$

$\sin\theta = \sqrt{\frac{1-\cos(2\theta)}{2}}$

Half Angle

$2\theta = \alpha$ (माना)

$\theta = \alpha/2$

$\cos(\alpha) = \cos^2(\alpha/2) - \sin^2(\alpha/2)$

Find Value (Comp) of :-

- $\sin(120) = \sin(180-60) = \sin 180 \times \cos 60 - \sin 60 \cos 180 = -\frac{\sqrt{3}}{2}(-1) = +\sqrt{3}/2$

(1) $\sin(120^\circ) = \sin(90+30)$

$= \sin 90 \cos 30 + \sin 30 \cos 90$

$= 1 \times \frac{\sqrt{3}}{2}$

$= \frac{\sqrt{3}}{2}$

$\sin(2 \times 60) = 2 \sin 60 \cdot \cos 60$

$= 2 \times \frac{\sqrt{3}}{2} \times \frac{1}{2} = \frac{\sqrt{3}}{2}$

(2) $\cos(150^\circ) = \cos(90+60)$

$= \cos 90 \cdot \cos 60 - \sin 90 \sin 60$

$= 0 \times -1 \cdot \frac{\sqrt{3}}{2} = -\frac{\sqrt{3}}{2}$

(3) $\sin(240^\circ)$

(4) $\cos(120^\circ) = \cos(90+30)$

$= -\frac{1}{2}$

(5) $\sin(90-\theta) = ??$ (cop, $\angle A$, $\angle \theta$)

$\sin(90-\theta) = \sin 90 \cos\theta - \sin\theta \cos 90$ (0)

$= 1 \times \cos\theta - 0$

$* \boxed{\sin(90-\theta) = \cos\theta}$

(6) $\sin(90+\theta) = \sin 90 \cdot \cos\theta + \sin\theta \cos 90$ (0)

$= \cos\theta$

(7) $\sin(180-\theta) = \sin 180 \cos\theta - \sin\theta \cdot \cos 180$

$= 0 - \sin\theta(-1)$

$= +\sin\theta$

(8) $\sin(270+\theta) = \sin 270 \cos\theta + \sin\theta \cos(270)$ (0)

$= -1 \cos\theta$

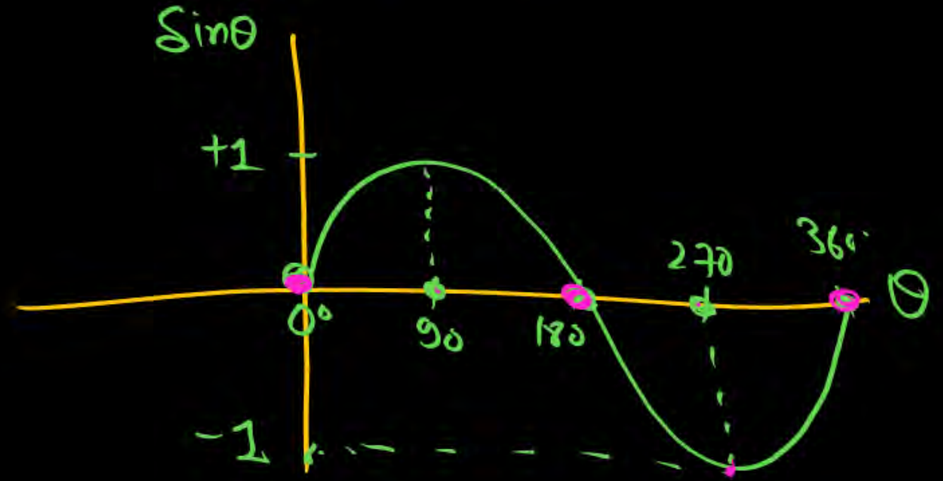
$= -\cos\theta$ ✓

Cop

$\sin(0^\circ) = 0 = \sin(180^\circ) = \sin(360^\circ)$

$\sin(90^\circ) = +1$

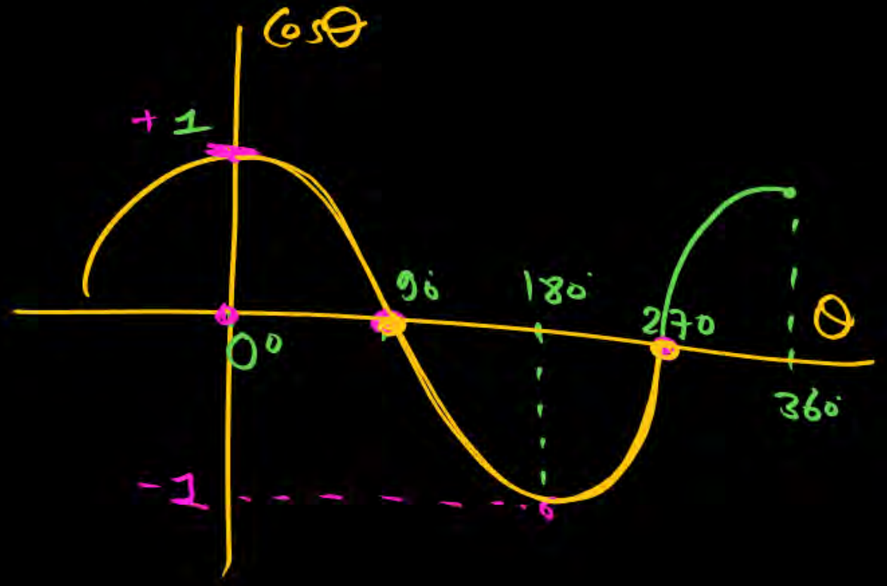
$\sin(270^\circ) = -1$



$\cos(0^\circ) = 1$

$\cos(180^\circ) = -1$

$\cos 90^\circ = 0$
$\cos(270^\circ) = 0$



COPY

(9) $\cos(360^\circ - \theta) = \cos 360^\circ \cdot \cos\theta + \sin 360^\circ \sin\theta$

$= 1 \times \cos\theta + 0$

$\cos(360 - \theta) = \cos\theta$

$\cos(0 - \theta) = \cos\theta$

$\boxed{\cos(-\theta) = \cos\theta}$ ← even function

Ex $\cos(-30) = \cos 30 = \frac{\sqrt{3}}{2}$

(10) $\sin(360^\circ - \theta) = \sin 360^\circ \cos\theta - \sin\theta \cos 360$

$= -\sin\theta \times 1$

$\sin(360 - \theta) = -\sin\theta$

$\boxed{\sin(-\theta) = -\sin\theta}$ * (odd fn.)

$\boxed{\tan(-\theta) = -\tan\theta}$

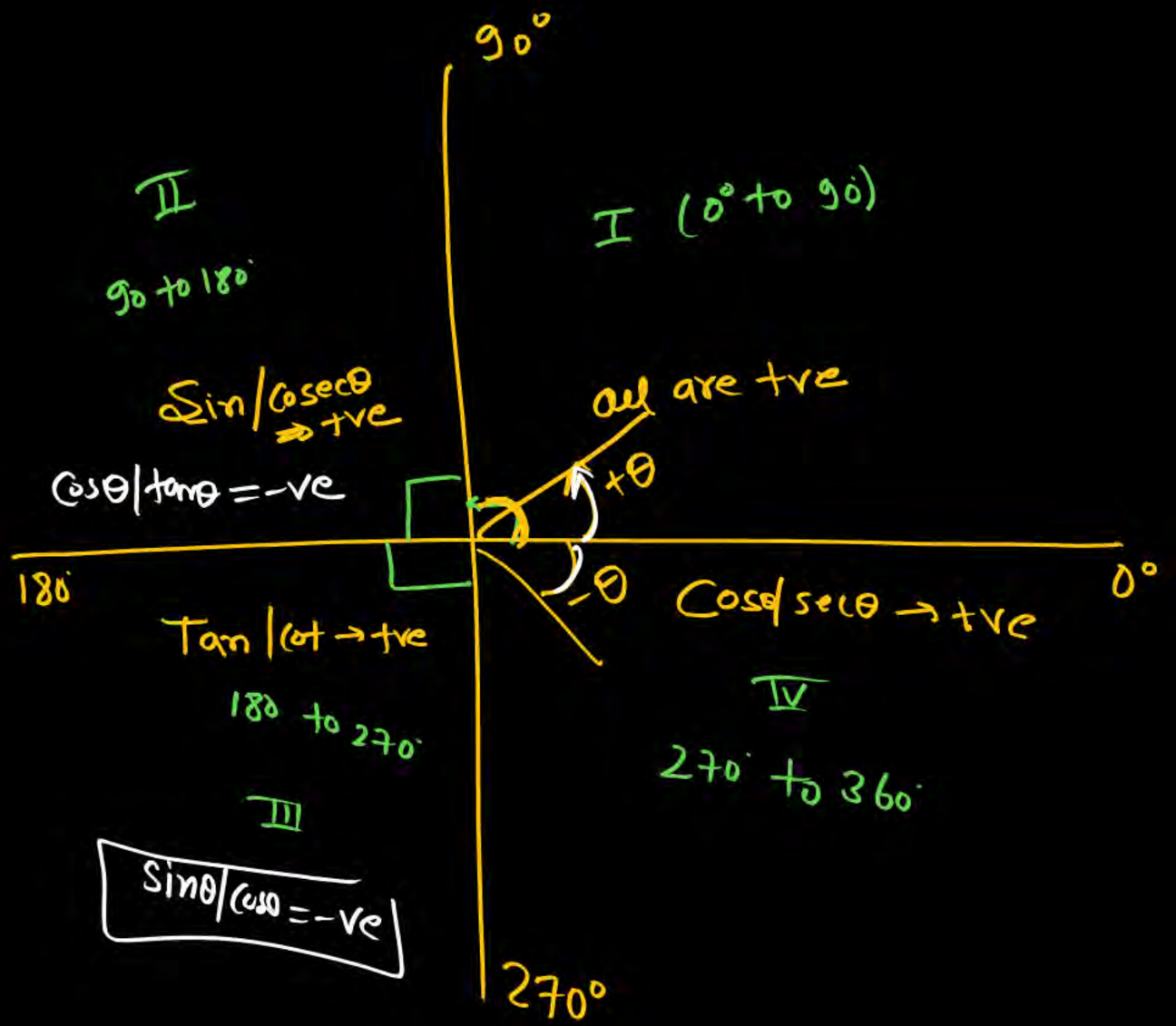
(11) $\sin(360 + \theta) = \sin 360^\circ \times \cos\theta + \cos 360 \sin\theta$

$\sin(360 + \theta) = \sin\theta$

(12) $\tan(150^\circ)$

$\tan(180 - 30) =$

Col



MR* BOX

$\cos(-\theta) = +\cos\theta$

$\sin(-\theta) = -\sin\theta$

$\tan(-\theta) = -\tan\theta$

Hw

COPY

(13) $\sin(180-\theta)$

(14) $\cos(180+\theta)$

cop

(#) $\sin(127^\circ) = \sin(90+37^\circ)$

$\tan(143^\circ) = \tan(180-37^\circ)$

$\cos(143^\circ) = \cos(90+53^\circ)$

Cop

$\sin(105^\circ) = \sin(60+45)$

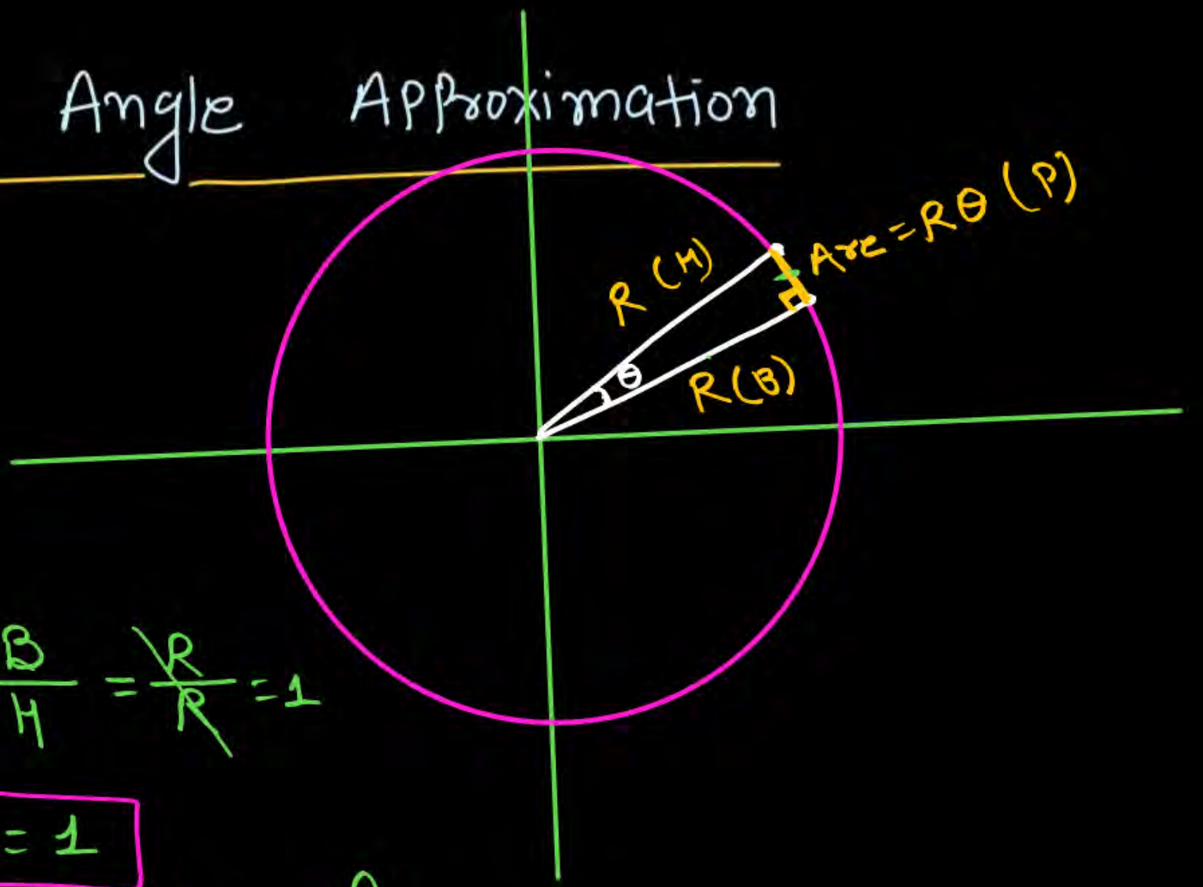
$=$

$\sin(75^\circ) = \sin(45^\circ+30^\circ)$

$\sin(15^\circ) = \sin(45^\circ-30^\circ) =$

Copy

Small Angle Approximation



$$\cos\theta = \frac{B}{H} = \frac{R}{R} = 1$$

\# $\boxed{\cos\theta = 1}$

only when θ is small.

$\theta \leq 5^\circ$

$$\sin\theta = \frac{P}{H} = \frac{R\theta}{R}$$

$\boxed{\sin\theta = \theta}$

must be in radian

$$\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{\theta}{1}$$

$\theta \to$ small

$\boxed{\tan\theta = \theta}$

Question

copy

Find value:

(i) $\sin 2^\circ = 2^\circ = \frac{2 \times \pi \text{ rad}}{180} = \frac{\pi}{90} \text{ rad}$

(ii) $\tan 3^\circ = \frac{3 \times \pi}{180} = \frac{\pi}{60} \text{ rad}$

(iii) $\cos 3^\circ = 1$

Copy

$\sin(-30°) =$

$\cos(-60°) =$

$\sin(-120°) =$

$\cos(-120°) =$

$\sin(-150°) =$

Question

Which of the following option is correct for the value of tan (2°)

1. 2°
2. π/90
3. π/30
4. 2 rad

$$\tan(\theta) = \theta$$

$\tan(2^\circ) = 2^\circ$

Question

H/W No copy

Suggest suitable match between function given in the first column and its description given in the second column.

1. A→PT, B →QT, C→QT, D→PS
2. A→PT, B →QS, C→QT, D→PS
3. A→QT, B →QS, C→PT, D→PS
4. A→QS, B →PT, C→QT, D→PS

Column-I		Column-II	
(A)	sin (390°)	(P)	Positive
(B)	sin (–30°)	(Q)	Negative
(C)	cos 120°	(R)	Zero
(D)	tan (–120°)	(S)	Modulus is greater than one
		(T)	Modulus is less than one



Telegram chu

← Join.

H/W

(1) Do DPP.

(2) 1st & 2nd lecture revision

(3) Sanghush Assignment-1 event me aayga

Thank You